Evaluation Cloud Computing

J'ai utilisé différentes ressources, premièrement pour télécharger tout ce qu'il fallait sur mon ordinateur personnel. Ensuite, les documentations fournies en cours. Pour finir, j'ai également utilisé chatgpt.

1) General questions

Q1. What is cloud computing?

➔ b. A model for on-demand delivery of IT resources over the internet

Q2. Which of the following is NOT a characteristic of cloud computing?

➔ c. High initial investment in hardware

Q3. What does the term "scalability" refer to in cloud computing?

➔ a. Increasing or decreasing resources as needed

Q4. Which Azure region is closer to users located in Western Europe?

➔ c. North Europe

Q5. What does "pay-as-you-go" mean in Azure?

➔ b. You only pay for what you use

Q6. What is Azure Virtual Machine (VM) used for?

➔ b. Hosting virtual servers in the cloud

Q7. What is Azure Blob Storage typically used for?

➔ a. Storing unstructured data, such as images, audio, and video

Q8. How does Azure handle data redundancy to ensure availability?

➔ b. Data is replicated across multiple locations or zones

Q9. Which of the following can help reduce Azure costs?

➔ a. Using Reserved Instances for predictable workloads

Q10. Which Azure service can be used to securely store secrets and access keys?

➔ c. Azure Key Vault

Q11. Which cloud deployment model provides the most control over resources but requires significant expertise to maintain?

➔ b. Private Cloud

Q12. Which Azure service provides a managed environment for serverless computing, where users don't manage infrastructure directly?

➔ b. Azure Functions

Q13. Which of the following services allows for the orchestration of multi-step processes, often connecting multiple Azure services in automated workflows?

➔ b. Azure Logic Apps

2) General questions

1) Pour se connecter à Azure : az login, il nous est demandé de choisir ensuite l'abonnement de notre choix, ici nous allons choisir le « Azure for students ».

```
ilyna@ilyna: ~
 authenticate.

Retrieving tenants and subscriptions for the selection...

[Tenant and subscription selection]

No     Subscription name    Subscription ID                       Tenant
-----  -------------------  ------------------------------------  -------------
[1] *  Azure for Students   c5559cae-a684-4ea1-a92a-cc5873f415e1  univ-lyon2.fr

The default is marked with an *; the default tenant is 'univ-lyon2.fr' and subscription is 'Azure for Students'
(c5559cae-a684-4ea1-a92a-cc5873f415e1).

Select a subscription and tenant (Type a number or Enter for no changes): 1

Tenant: univ-lyon2.fr
Subscription: Azure for Students (c5559cae-a684-4ea1-a92a-cc5873f415e1)

[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more
 about it and its configuration at https://go.microsoft.com/fwlink/?linkid=2271236

If you encounter any problem, please open an issue at https://aka.ms/azclibug

[Warning] The login output has been updated. Please be aware that it no longer displays the full list of availab
le subscriptions by default.
```

2) az group create --name cinearchive-ilm-rg --location francecentral → pour créer le groupe de ressource avec les deux premières lettres de mon prénom et la première lettre de mon nom de famille.

```
ilyna@ilyna:~$ az group create --name cinearchive-ilm-rg --location francecentral
{
  "id": "/subscriptions/c5559cae-a684-4ea1-a92a-cc5873f415e1/resourceGroups/cinearchive-ilm-rg",
  "location": "francecentral",
  "managedBy": null,
  "name": "cinearchive-ilm-rg",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
```

az storage account create --name cinearchiveilm --resource-group cinearchive-ilm-rg --location francecentral --sku Standard_GRS → on va créer le storage account.

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```
ilyna@ilyna: $ az storage account create --name cinearchiveilm --resource-group cinearchive-ilm-rg --location francecentral --sku Standard_GRS
{
  "accessTier": "Hot",
  "accountMigrationInProgress": null,
  "allowBlobPublicAccess": false,
  "allowCrossTenantReplication": false,
  "allowSharedKeyAccess": null,
  "allowedCopyScope": null,
  "azureFilesIdentityBasedAuthentication": null,
  "blobRestoreStatus": null,
  "creationTime": "2024-11-07T15:28:10.850757+00:00",
  "customDomain": null,
  "defaultToOAuthAuthentication": null,
  "dnsEndpointType": null,
  "enableExtendedGroups": null,
  "enableHttpsTrafficOnly": true,
  "enableNfsV3": null,
  "encryption": {
    "encryptionIdentity": null,
    "keySource": "Microsoft.Storage",
    "keyVaultProperties": null,
    "requireInfrastructureEncryption": null,
    "services": {
      "blob": {
        "enabled": true,
        "keyType": "Account",
        "lastEnabledTime": "2024-11-07T15:28:11.163262+00:00"
      },
      "file": {
        "enabled": true,
        "keyType": "Account",
        "lastEnabledTime": "2024-11-07T15:28:11.163262+00:00"
      },
      "queue": null,
      "table": null
    }
  },
  "extendedLocation": null,
  "failoverInProgress": null,
  "geoReplicationStats": null,
  "id": "/subscriptions/c5559cae-a684-4ea1-a92a-cc5873f415e1/resourceGroups/cinearchive-ilm-rg/providers/Microsoft.Storage/storageAccounts/cinearchiveilm",
  "identity": null,
  "immutableStorageWithVersioning": null,
  "isHnsEnabled": null,
  "isLocalUserEnabled": null,
  "isSftpEnabled": null,
  "isSkuConversionBlocked": null,
  "keyCreationTime": {
    "key1": "2024-11-07T15:28:10.928891+00:00",
    "key2": "2024-11-07T15:28:10.928891+00:00"
  },
  "keyPolicy": null,
  "kind": "StorageV2",
  "largeFileSharesState": null,
  "lastGeoFailoverTime": null,
  "location": "francecentral",
  "minimumTlsVersion": "TLS1_0",
  "name": "cinearchiveilm",
  "networkRuleSet": {
    "bypass": "AzureServices",
    "defaultAction": "Allow",
    "ipRules": [],
    "ipv6Rules": [],
    "resourceAccessRules": null,
    "virtualNetworkRules": []
  },
  "primaryEndpoints": {
    "blob": "https://cinearchiveilm.blob.core.windows.net/",
    "dfs": "https://cinearchiveilm.dfs.core.windows.net/",
    "file": "https://cinearchiveilm.file.core.windows.net/",
    "internetEndpoints": null,
    "microsoftEndpoints": null,
    "queue": "https://cinearchiveilm.queue.core.windows.net/",
    "table": "https://cinearchiveilm.table.core.windows.net/",
    "web": "https://cinearchiveilm.z28.web.core.windows.net/"
  },
  "primaryLocation": "francecentral",
  "privateEndpointConnections": [],
  "provisioningState": "Succeeded",
  "publicNetworkAccess": null,
  "resourceGroup": "cinearchive-ilm-rg",
  "routingPreference": null,
  "sasPolicy": null,
  "secondaryEndpoints": null,
  "secondaryLocation": "francesouth",
  "sku": {
    "name": "Standard_GRS",
    "tier": "Standard"
  },
  "statusOfPrimary": "available",
  "statusOfSecondary": "available",
  "storageAccountSkuConversionStatus": null,
  "tags": {},
  "type": "Microsoft.Storage/storageAccounts"
}
ilyna@ilyna: $
```

GRS signifie **Geo-Redundant Storage**, qui est un type de stockage permettant la réplication géographique des données. (chatgpt) Nous avons choisi l'option **Standard_GRS**, ce qui signifie que le stockage sera répliqué dans une autre région (sert en cas de panne).

3) az storage container create --name raw-images-ilm-test --account-name cinearchiveilm --account-key "..." → créer le conteneur raw image.
   az storage container create --name edited-images-ilm-test --account-name cinearchiveilm --account-key "..."→ créer le conteneur edited image.

J'ai essayé deux méthodes, d'abord avec «--auth-mode login » à la fin mais cela généré une erreur alors j'ai remplacé par « --account-key » et sa valeur.

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```
ilyna@ilyna:~$ az storage container create --name raw-images-ilm
-test --account-name cinearchiveilm --account-key "VSgAUVbiltTyY
Rm1mSJtR7JhNdKXA4d68hNVs3CZWnBDbXhbubObspdu30r/eA4rhDqzSNTh8wEn+
AStHAI88Q=="
az storage container create --name edited-images-ilm-test --acco
unt-name cinearchiveilm --account-key "VSgAUVbiltTyYRm1mSJtR7JhN
dKXA4d68hNVs3CZWnBDbXhbubObspdu30r/eA4rhDqzSNTh8wEn+AStHAI88Q=="

{
  "created": true
}
{
  "created": true
}
ilyna@ilyna:~$
```

4) URL du compte de stockage : https://cinearchive-ilm.blob.core.windows.net
   URL du conteneur 1 :https://cinearchive-ilm.blob.core.windows.net/raw-images-ilm
   URL du conteneur 2 : https://cinearchive-ilm.blob.core.windows.net/edited-images-ilm
5) az storage blob upload-batch --account-name cinearchiveilm --destination raw-images-ilm --source /mnt/c/Users/ilyna/OneDrive/Bureau/cloud/evaluation-cc-m1/
6) az storage blob list --account-name cinearchiveilm --container-name raw-images-ilm --output table

```
Name                                                  Blob Type    Blob Tier    Length    Content Type    Last Modified
     Snapshot
----------------------------------------------------  -----------  -----------  --------  --------------  ------------------------
----------
codioful-formerly-gradienta-bKESVqfxass-unsplash.jpg  BlockBlob    Hot          455605    image/jpeg      2024-11-07T16:18:32+00:00
david-becker-dMeEJRE18VI-unsplash.jpg                 BlockBlob    Hot          867052    image/jpeg      2024-11-07T16:18:32+00:00
jr-korpa-E2i7Hftb_rI-unsplash.jpg                     BlockBlob    Hot          1002894   image/jpeg      2024-11-07T16:18:32+00:00
mohammad-alizade-XgeZu2jBaVI-unsplash.jpg             BlockBlob    Hot          536794    image/jpeg      2024-11-07T16:18:33+00:00
pawel-czerwinski-bX9B9c-YasY-unsplash.jpg             BlockBlob    Hot          2174474   image/jpeg      2024-11-07T16:18:33+00:00
pawel-czerwinski-prMn9KINLtI-unsplash.jpg             BlockBlob    Hot          2333038   image/jpeg      2024-11-07T16:18:34+00:00
pexels-felix-mittermeier-956981.jpg                   BlockBlob    Hot          9751220   image/jpeg      2024-11-07T16:18:36+00:00
pexels-louis-2101187.jpg                              BlockBlob    Hot          1455177   image/jpeg      2024-11-07T16:18:37+00:00
pexels-philippe-donn-1169754.jpg                      BlockBlob    Hot          3097667   image/jpeg      2024-11-07T16:18:37+00:00
rodion-kutsaiev-kE0mU-yy31Y-unsplash.jpg              BlockBlob    Hot          4226515   image/jpeg      2024-11-07T16:18:38+00:00
tengyart-mw8GlS4bymY-unsplash.jpg                     BlockBlob    Hot          1112559   image/jpeg      2024-11-07T16:18:39+00:00
ilyna@ilyna:~$
```

7) az storage blob copy start-batch --account-name cinearchiveilm --source-container raw-images-ilm --destination-container edited-images-ilm

```
[
  "https://cinearchiveilm.blob.core.windows.net/edited-images-il
m/codioful-formerly-gradienta-bKESVqfxass-unsplash.jpg",
  "https://cinearchiveilm.blob.core.windows.net/edited-images-il
m/david-becker-dMeEJRE18VI-unsplash.jpg",
  "https://cinearchiveilm.blob.core.windows.net/edited-images-il
m/jr-korpa-E2i7Hftb_rI-unsplash.jpg",
  "https://cinearchiveilm.blob.core.windows.net/edited-images-il
m/mohammad-alizade-XgeZu2jBaVI-unsplash.jpg",
  "https://cinearchiveilm.blob.core.windows.net/edited-images-il
m/pawel-czerwinski-bX9B9c-YasY-unsplash.jpg",
  "https://cinearchiveilm.blob.core.windows.net/edited-images-il
m/pawel-czerwinski-prMn9KINLtI-unsplash.jpg",
  "https://cinearchiveilm.blob.core.windows.net/edited-images-il
m/pexels-felix-mittermeier-956981.jpg",
  "https://cinearchiveilm.blob.core.windows.net/edited-images-il
m/pexels-louis-2101187.jpg",
  "https://cinearchiveilm.blob.core.windows.net/edited-images-il
m/pexels-philippe-donn-1169754.jpg",
  "https://cinearchiveilm.blob.core.windows.net/edited-images-il
m/rodion-kutsaiev-kE0mU-yy31Y-unsplash.jpg",
  "https://cinearchiveilm.blob.core.windows.net/edited-images-il
m/tengyart-mw8GlS4bymY-unsplash.jpg"
]
ilyna@ilyna:~$
```


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8) az storage blob list --account-name cinearchiveilm --container-name edited-images-ilm --output table

```
Name                                                Blob Type    Blob Tier    Length    Content Type    Last Modified
     Snapshot
--------------------------------------------------  -----------  -----------  --------  --------------  ------------------------
----------
codioful-formerly-gradienta-bKESVqfxass-unsplash.jpg  BlockBlob    Hot          455605    image/jpeg      2024-11-07T16:23:53+00:00
david-becker-dMeEJRE18VI-unsplash.jpg               BlockBlob    Hot          867052    image/jpeg      2024-11-07T16:23:53+00:00
jr-korpa-E2i7Hftb_rI-unsplash.jpg                   BlockBlob    Hot          1002894   image/jpeg      2024-11-07T16:23:53+00:00
mohammad-alizade-XgeZu2jBaVI-unsplash.jpg           BlockBlob    Hot          536794    image/jpeg      2024-11-07T16:23:54+00:00
pawel-czerwinski-bX9B9c-YasY-unsplash.jpg           BlockBlob    Hot          21744474  image/jpeg      2024-11-07T16:23:54+00:00
pawel-czerwinski-prMn9KINLtI-unsplash.jpg           BlockBlob    Hot          2333038   image/jpeg      2024-11-07T16:23:54+00:00
pexels-felix-mittermeier-956981.jpg                 BlockBlob    Hot          9751220   image/jpeg      2024-11-07T16:23:54+00:00
pexels-louis-2101187.jpg                            BlockBlob    Hot          1455177   image/jpeg      2024-11-07T16:23:54+00:00
pexels-philippe-donn-1169754.jpg                    BlockBlob    Hot          3097667   image/jpeg      2024-11-07T16:23:54+00:00
rodion-kutsaiev-kE0mU-yy31Y-unsplash.jpg            BlockBlob    Hot          4226515   image/jpeg      2024-11-07T16:23:54+00:00
tengyart-mw8GlS4bymY-unsplash.jpg                   BlockBlob    Hot          1112559   image/jpeg      2024-11-07T16:23:54+00:00
ilyna@ilyna:~$
```

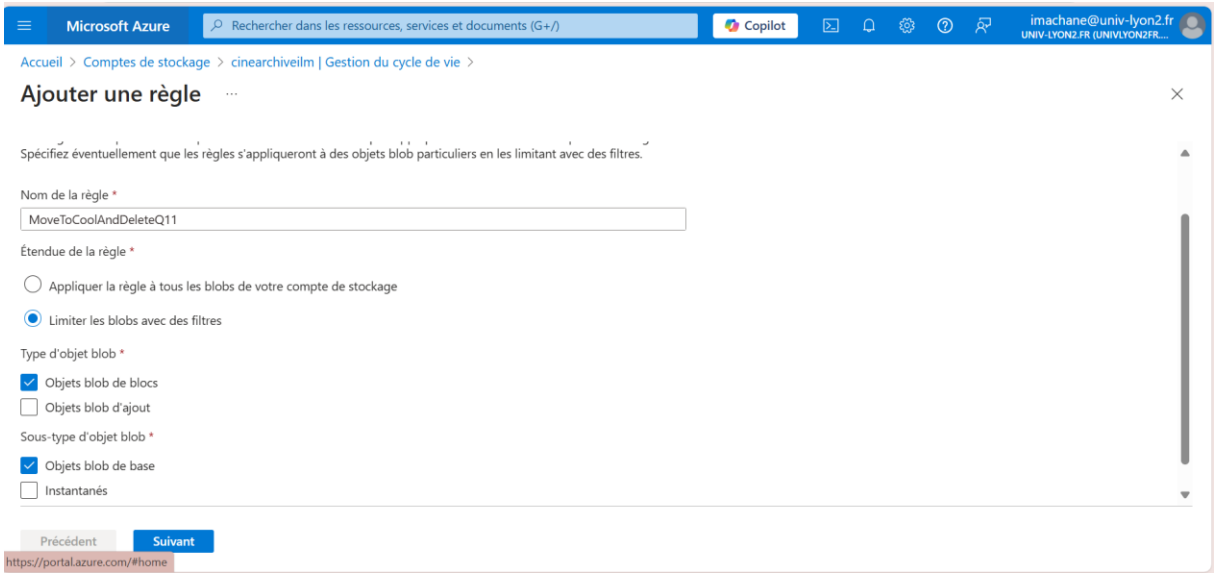
9) az storage blob delete-batch --account-name cinearchiveilm --source raw-images-ilm --pattern "*"

10) az storage management-policy rule create \
 --account-name cinearchiveilm \
 --container-name raw-images-ilm \
 --rule-name "MoveToCoolAndDelete" \
 --policy '{"rules": [{"filters": {"blobTypes": ["blockBlob"]}, "actions": {"baseBlob": {"modify": [{"tierToCool": {"daysAfterLastModificationGreaterThan": 60}}, {"delete": {"daysAfterLastModificationGreaterThan": 365}}]}}}]}'

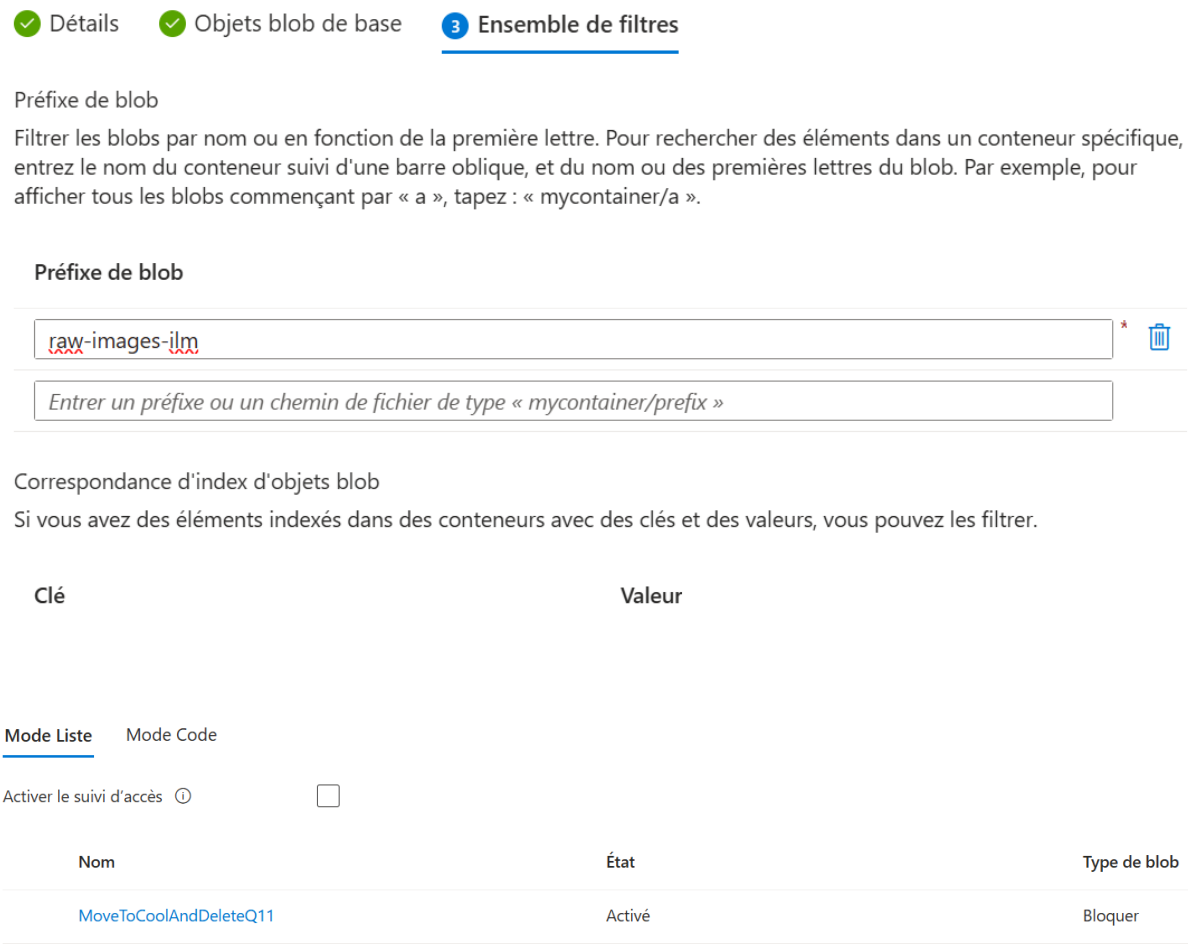
```
{
 "rules": [
  {
   "filters": {
    "blobTypes": ["blockBlob"]
   },
   "actions": {
    "baseBlob": {
     "modify": [
      {
       "tierToCool": {
        "daysAfterLastModificationGreaterThan": 60
       }
      },
      {
       "delete": {
        "daysAfterLastModificationGreaterThan": 365
       }
      }
     ]
    }
```

```
        }
      }
    ]
  }
```

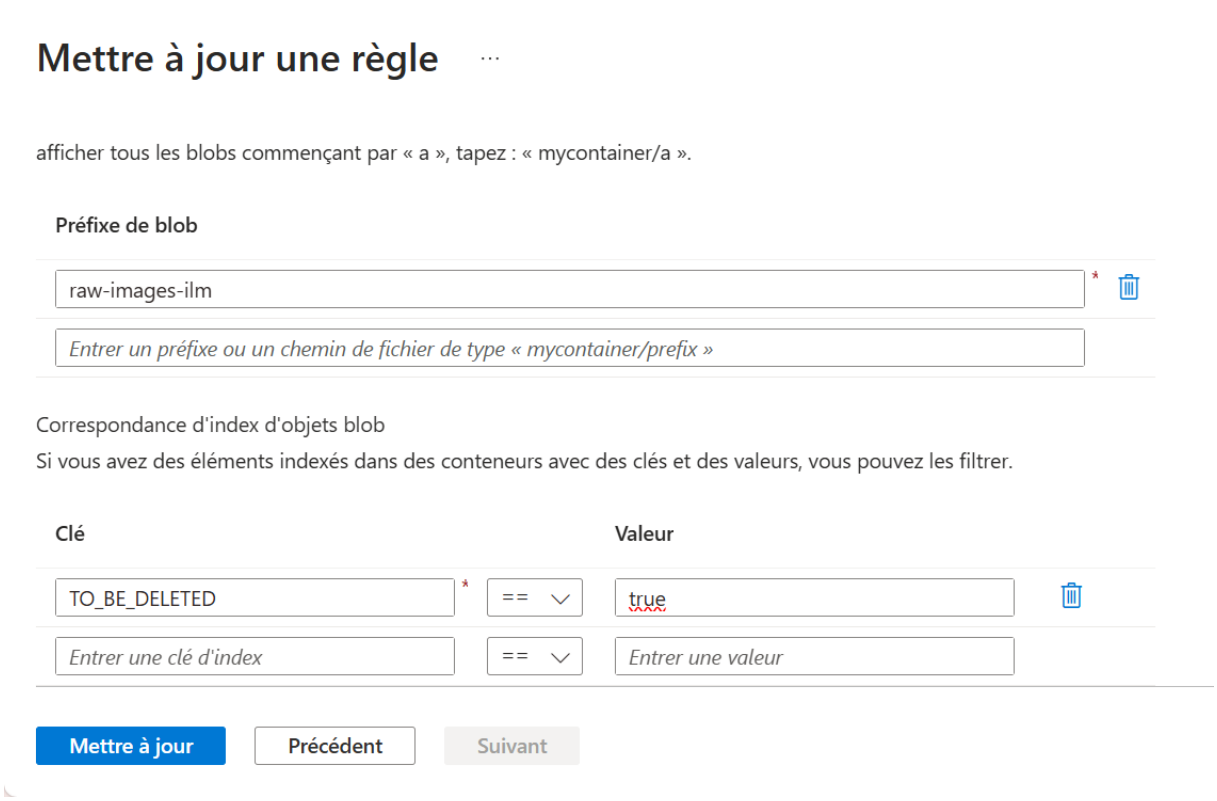
11)


Microsoft Azure
Rechercher dans les ressources, services et documents (G+/)
Copilot
imachane@univ-lyon2.fr
UNIV-LYON2.FR (UNIVLYON2FR....
Accueil > Comptes de stockage > cinearchiveilm | Gestion du cycle de vie >
Ajouter une règle
Spécifiez éventuellement que les règles s'appliqueront à des objets blob particuliers en les limitant avec des filtres.
Nom de la règle *
MoveToCoolAndDeleteQ11
Étendue de la règle *
Appliquer la règle à tous les blobs de votre compte de stockage
Limiter les blobs avec des filtres
Type d'objet blob *
Objets blob de blocs
Objets blob d'ajout
Sous-type d'objet blob *
Objets blob de base
Instantanés
Précédent
Suivant
https://portal.azure.com/#home

Si
Objets blob de base ont été *
Modifiés pour la dernière fois
Créé
Il y a plus de (jours) *
60
Alors
Déplacer vers le stockage froid

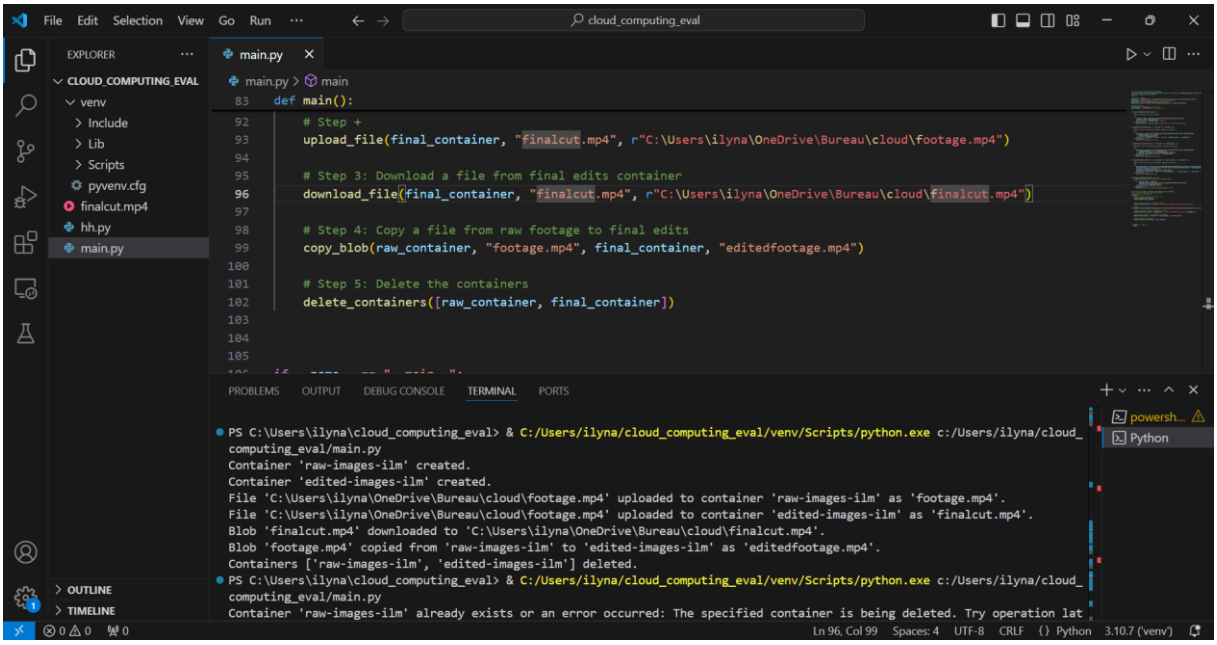


12) Pour nous allons juste ajouter un tag à ce qui a été fait précédemment.



13) az storage container delete --account-name cinearchiveilm --name raw-images-ilm

az storage container delete --account-name cinearchiveilm --name edited-images-ilm

On peut même vérifier si le storage account ne contient rien avec : az storage container list --account-name cinearchiveilm --output table

3) Python code